

REVIEW

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Cytomegalovirus infection in day care centres: A systematic review and meta-analysis of prevalence of infection in children

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Summary

Maternofetal transmission of cytomegalovirus (CMV) is the most common infectious cause of congenital malformation in developed countries. Maternal infection often results from close contact with infected children, and this may occur in day care centres (DCCs).

A systematic review of observational studies was conducted to examine the prevalence of CMV infection among children attending DCCs. Meta-analysis using the random effect model was performed for studies including controls. Sources included PubMed, EMBASE (until August 2018), and references from identified publications. Inclusion criteria were studies reporting CMV infection prevalence among childcare children aged less than 7 years of age. Controls were children without childcare exposure. CMV infection was defined as viral excretion detected by culture, polymerase chain reaction, or CMV seropositivity.

Twenty-eight publications including 8347 participants met the eligibility criteria. The pooled prevalence of CMV infection among children in childcare from all studies was 32% (95% CI 23–41). Within case-controlled studies, prevalence among children attending DCCs was 34% (95% CI 25–44), whereas prevalence among those without childcare exposure was 22% (95% CI 15–30). Meta-analysis showed a significant association between DCC attendance and CMV infection (odds ratio 2.69, 95% CI 1.68–4.30; heterogeneity $\chi^2/\text{df} = 8$; $I^2 = 84\%$, $P < 0.00001$).

Attendance at DCCs is significantly associated with increased risk of childhood CMV infection. Prevention strategies to reduce risk of CMV infection of pregnant women and children should involve review of DCC exposure and consideration of preventative hygiene strategies.

KEYWORDS

cytomegalovirus infection, day care centres, epidemiology, prevalence, viral transmission

1 | INTRODUCTION

Human cytomegalovirus (CMV) is the major infectious cause of malformation of newborns in developed countries.^{1,2} Similarly, high rates of

CMV-induced malformation may occur in developing countries, although transmission data are only available for a small number of these countries.³ The estimated mean prevalence of CMV infection in utero (congenital CMV infection) is 0.64%, with considerable variation worldwide.² Congenital CMV infection can result in fetal or neonatal death, and permanent disabilities in children, including sensorineural hearing loss, blindness, mental disability, and neurodevelopmental delay.⁴

Abbreviations: CI, Confidence interval; DCC, Day care centre; HC, Home care; NOS, Newcastle-Ottawa Scale; OR, Odds ratio

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Seropositivity for CMV, which identifies individuals who have acquired CMV infection, is found in 45 to >90% of women of reproductive age ranges, and varies between countries, socio-economic status, and ethnicity.^{5,6} A primary CMV infection occurs annually in approximately 1% to 2% of seronegative pregnant women.⁷ However, both a primary and secondary maternal CMV infection (ie, reactivation of latent virus or re-infection with a different strain) can lead to maternofetal transmission of CMV, causing disease in the infant, and significant burden of disease worldwide.⁸⁻¹⁰

Acquisition of CMV usually occurs via close contact with an individual excreting CMV in their body fluids, including saliva, urine, and blood.¹¹ In particular, saliva and urine of CMV-infected children are common sources of transmission to pregnant women.¹² CMV transmission is known to occur from young children attending a day care centre (DCC) to their siblings and parents.^{13,14} Research shows many mothers with a CMV-infected child less than 2 years of age and attending day care seroconvert within 1 year of commencement at day care, at a rate of 30% compared with 3% for those without infected children.¹⁵ Staff at DCCs who are repeatedly exposed to CMV, many of whom are women of childbearing age, are also at risk of CMV infection.^{16,17} Despite this risk, a large proportion (61% to 87.5%) of pregnant women and providers of their clinical care are unaware and uninformed about congenital CMV infection.¹⁸⁻²⁵

Given that transmission to pregnant women from infected children, including those in childcare, is a significant risk factor for CMV infection during pregnancy, it is essential to quantify this risk in order to (1) aid discussions with pregnant women about risk; (2) better define prognosis in at-risk women; (3) improve discussion with, and counselling of, pregnant women regarding prevention; and (4) inform policy discussions regarding prevention methods for congenital CMV. We systematically reviewed the prevalence of CMV infection in children attending DCCs and analysed the factors contributing to CMV infection among these children. Such data can be used to inform strategies to prevent CMV transmission to adults in childcare and allow better information for pregnant women and their carers.

2 | METHODS

Two reviewers independently conducted a systematic search for studies of CMV infection in children attending DCCs. Databases searched were PubMed and EMBASE from 1947 until August 2018. Search terms for Pubmed were ((cytomegalovirus infection [MeSH Terms]) AND (((child care [MeSH Terms]) OR day care [MeSH Terms]) OR child day care centers [MeSH Terms]) OR preschool) OR nursery)) AND (((disease transmission, infectious [MeSH Terms]) OR viral shedding [MeSH Terms]) OR prevalence) OR molecular epidemiology [MeSH Terms]) OR epidemiology). The search strategy for EMBASE is shown in Supporting Information Table S1. The search was limited to studies in humans and studies in English and was supplemented by hand searching reference lists of the identified papers.

Studies were eligible for inclusion if (1) they were case-control or uncontrolled cohort/cross sectional studies; (2) the study population was of pre-school age (defined as less than 7 years); (3) subjects attended childcare facilities that were DCCs and day nurseries; (4)

the study controls had no childcare exposure; and (5) the prevalence of CMV infection was reported as either CMV excretion detected by culture or polymerase chain reaction (PCR) or seropositivity detected by the presence of CMV antibodies. For prospective studies, presence of CMV virus excretion at the end of the study was used. Studies were ineligible for inclusion if they were (1) review articles; (2) case reports; (3) letters to the editor; (4) articles not published in peer-reviewed journals; and (5) the study subjects attended non-DCC settings, including hospital nurseries and orphanages.

All studies were critically appraised using the Newcastle-Ottawa Scale (NOS).²⁶ Modifications were made to enhance the suitability of the NOS in different study types. The original NOS was used for case-controlled studies (Supporting Information Table S2 and Supporting Information Figure S1) with a total score of nine awarded to three areas of evaluation: selection, comparability, and outcome. A score of five or more was considered to represent moderate to good study quality. In the comparability category, studies were evaluated if they had controlled for age, which is the most important factor to influence CMV prevalence. For the uncontrolled studies, the same three categories of evaluation were employed, with some criteria deemed inapplicable and omitted (Figures S2 and S3). As such, the prospective cohort (Supporting Information Table S3) and cross-sectional studies (Supporting Information Table S4) were evaluated relative to total scores of eight and six, respectively. A score of >50% of the maximum score was considered as moderate to good study quality.

For controlled studies, unadjusted odds ratios (OR) with 95% CI and *P* values for CMV infection among DCC children versus those in the control group were calculated using Mantel-Haenszel method with random effects models. *I*² statistics and χ^2/df were employed to determine statistical heterogeneity. Due to significant heterogeneity among studies, only results from random effects models are presented. Pooled prevalence of CMV infection (with 95% CI) were reported separately for controlled and uncontrolled studies, and as a combined estimate. Standard error (SE) was calculated using the following formula: $SE = \sqrt{p(1 - p)/n}$; where *p* = prevalence and *n* = number of cases or controls. Analyses were performed in Review Manager (v5.3, Copenhagen, The Cochrane Collaboration).

Data were extracted on variables that are putative risk factors for CMV infection among children attending DCCs, including age, age at first DCC attendance, DCC size, close proximity, duration of DCC attendance, and breastfeeding.

3 | RESULTS

The primary search yielded 894 publications and abstracts. After review of titles and abstracts regarding suitability for inclusion, 27/894 studies were included, with one additional study identified from reference lists (Figure 1). Of the 28 included studies, involving a total study population of 8347 children, 9/28 were case-control studies with 1824 children attending DCCs and 3477 controls (Table 1).²⁷⁻³⁵ Of these, 1/9 publications also reported two uncontrolled cross-sectional studies involving 20 and 40 subjects, respectively (Table 2).²⁷ Within the remaining 19/28 uncontrolled studies involving 3046 subjects, 11/19 were prospective cohort studies with a study period ranging from 2 months to 4.5 years, and 8/19 were cross-sectional studies (Table 2).³⁶⁻⁵⁴

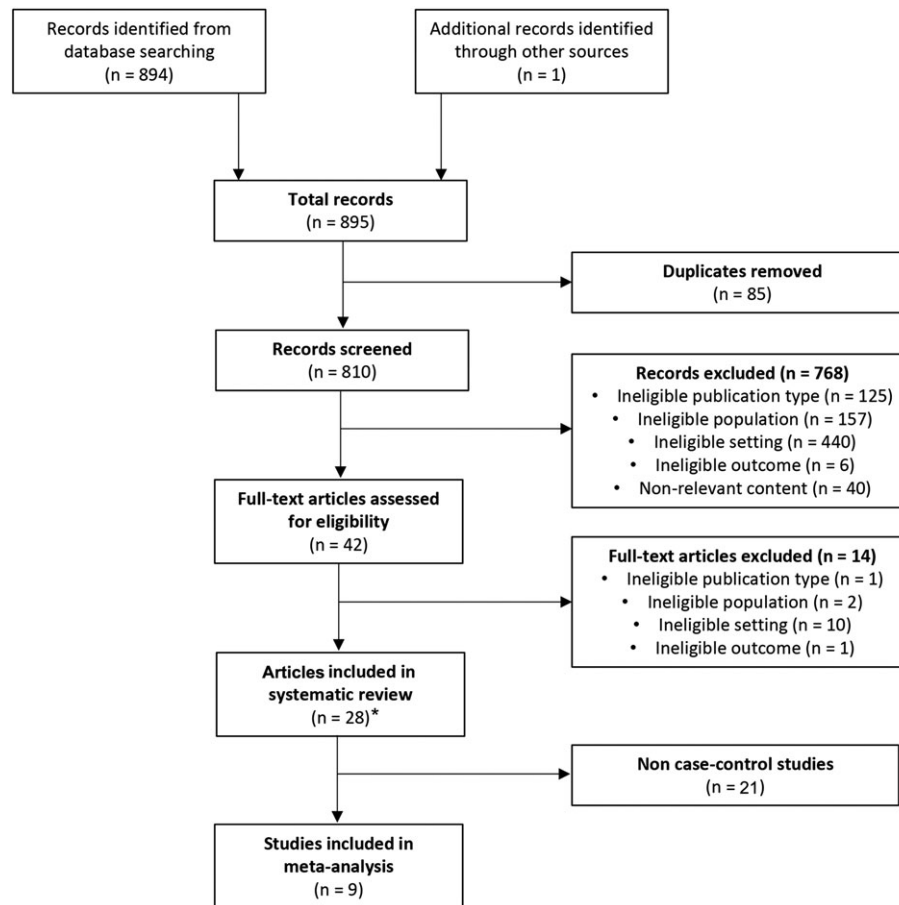


FIGURE 1 Search strategy for the identification of studies reporting the prevalence of CMV excretion among children attending DCCs. *One of the 28 articles reported a case-control study and two non case-controlled studies²⁷

TABLE 1 Prevalence of CMV excretion among children attending DCCs in controlled studies. All studies were evaluated using the Newcastle-Ottawa Scale (NOS)

Study	Location	Age (years)	Sample Type	No of DCCs	Method of Testing CMV in Children	CMV + ve/total (%)		NOS
						DCC	Control	
Strangert et al ^{27a}	Sweden	0.5-4	Serum	1	Serology	7/51 (14)	4/40 (10)	4/9
Pass et al ²⁸	USA	0-5	Urine, saliva, serum	1	Culture, serology	59/103 (57)	5/47 (11)	6/9
Adler et al ²⁹	USA	3-5 ^b	Urine	1	Culture	16/66 (25)	101/1457 (6.9)	5/9
Hutto et al ³⁰	USA	0.12-4	Urine, saliva, serum	3	Culture, serology	77/188 (41)	10/66 (15)	6/9
Blackman et al ³¹	USA	0.33-5	Urine, saliva	2	Culture	1/30 (3.3)	6/62 (9.8)	5/9
Liu et al ³²	China	0.08-7	Serum	N/A	Serology	543/823 (66)	602/1158 (52)	8/9
Kashiwagi et al ³³	Japan	0-2	Saliva	2	PCR	12/54 (22.2)	4/61 (6.5)	5/9
O'Brien et al ³⁴	New Zealand	0-3.5	Serum	N/A	Serology	86/253 (32.5)	71/217 (29)	6/9
Grosjean et al ³⁵	France	0.25-6	Saliva	6	PCR	133/256 (51.9)	80/369 (21.7)	8/9

^aPreliminary study section.

^bOne child was 8 years old.

Abbreviation: PCR, polymerase chain reaction.

3.1 | Characteristics of included studies

The participating children were predominantly from developed countries, with 14/28 studies conducted in North America and 8/28 in Europe (Tables 1 and 2). Three studies were conducted in developing countries—Brazil, China, and Mexico.^{32,51,54} Ten studies recruited children from a single centre and 18 studies involved multiple DCCs, with

between two and 38 included in a single study. Study populations varied in age distribution. In the case-control studies, control subjects attended home care in four studies,^{27,28,30,32} hospitals in three studies,^{29,31,35} and unspecified in two studies.^{33,34}

Cytomegalovirus was detected using viral culture followed by observation of cytopathic effect in 24/28 of the included studies, of

TABLE 2 Prevalence of CMV excretion among children attending DCCs in uncontrolled studies. All studies were evaluated using the Newcastle-Ottawa Scale (NOS)

Study	Location	Age (years)	Sample Type	No of DCCs	Methods of Testing CMV in Children	CMV + ve/total (%)	Study Type	NOS
Strangert et al ^{27a}	Sweden	0.58-2.5	Urine, serum	1	Culture, serology	7/20 (70)	Cross-sectional	3/6
Strangert et al ^{27b}	Sweden	0.5-3	Urine	1	Culture	9/40 (22.5)	Cross-sectional	3/6
Diosi et al ³⁶	Romania	0.25-3	Urine	1	Culture	5/30 (16.7)	Cross-sectional	3/6
Pass et al ³⁷	USA	0.25-5.4	Urine, saliva	1	Culture	36/70 (51)	Cross-sectional	3/6
MMWR ³⁸	USA	0->4	Urine, saliva	5	Culture	32/241 (13.3)	Cross-sectional	4/6
Jones et al ³⁹	USA	0.17-3	Urine, saliva	4	Culture	20/90 (22)	Cross-sectional	4/6
Hutto et al ⁴⁰	USA	0.12-5	Urine, saliva	1	Culture	22/55 (22)	Cross-sectional	4/6
Murph, et al ⁴¹	USA	0.25-6.1	Urine, saliva	1	Culture	9/41 (22)	Prospective cohort	6/8
Nelson et al ⁴²	UK	Not specified	Urine	5	Culture	32/117 (27)	Cross-sectional	5/6
Adler et al ⁴³	USA	0.17-3.3 ^c	Urine, saliva, serum ^d	1	Culture, serology	55/104 (53)	Prospective cohort	5/8
Murph and Bale ⁴⁴	USA	0.083-6.1	Urine, saliva	1	Culture	28/79 (35)	Prospective cohort	5/8
Volpi et al ⁴⁵	Italy	2.5	Saliva	9	Culture	33/253 (13)	Cross-sectional	5/6
Grillner and Strangert ⁴⁶	Sweden	1-7	Urine	2	Culture	13/31 (42)	Prospective cohort	5/8
Murph et al ⁴⁷	USA	0.038-5	Urine, saliva	3	Culture	32/219 (15)	Prospective cohort	5/8
Adler et al ⁴⁸	USA	< 3	Urine, serum ^c	3	Culture, serology	131/309 (42.4)	Prospective cohort	6/8
Ford-Jones et al ⁴⁹	Canada	0.25-3.5	Urine	38	Culture	79/471 (16.8)	Cross-sectional	4/6
Shen et al ⁵⁰	Taiwan	3-5	Urine, serum	2	Culture, serology, PCR	138/287 (48.1)	Prospective cohort	6/8
de Mello et al ⁵¹	Brazil	0.42-2.9	Urine, saliva, serum	1	Culture, serology	65/98 (66.3)	Prospective cohort	7/8
Lasry et al ⁵²	France	< 1	Urine	6	Culture, PCR	33/93 (35.5)	Prospective cohort	4/8
Chomel et al ⁵³	France	0.5-1	Urine	29	Culture	78/246 (31.7)	Prospective cohort	5/8
Noyola et al ⁵⁴	Mexico	< 6	Saliva	7	Culture	17/152 (11.2)	Prospective cohort	3/8

^aMain study section.^bSupplementary study section.^cMajority within this age group.^dSerum was collected from non-viruric children.

Abbreviation: PCR, polymerase chain reaction.

which most were published during or prior to 1996 (Tables 1 and 2). With the exception of O'Brien et al,³⁴ Chomel et al,⁵³ and Noyola et al⁵⁴ studies published after 1996 (4/28) employed molecular testing (PCR) to either confirm the cytopathology of culture results or as the primary and sole method for CMV detection. Liu et al and O'Brien et al³⁴ used serology alone to detect CMV infection.

The NOS analysis revealed good methodological quality, scoring 5 to 8 for 8/9 case-controlled studies, 4 to 7 for 10/11 prospective cohort studies, and 3 to 5 for all of the cross-sectional studies (Supporting Information Tables S2-S4). Study quality was mainly reduced by unreported non-response rate in case-control studies, inadequate follow-up in cohort studies, and absence of non-exposed cohort among cross-sectional studies.

3.2 | Prevalence of CMV infection among children attending childcare

The pooled prevalence of CMV infection among all children attending DCCs was 32% (95% CI 23%-41%). In the controlled studies, the pooled prevalence of CMV infection in children attending DCCs was 34% (95% CI 25%-44%) compared with 22% (95% CI 15-30) in the control (non-DCC) group.

3.3 | Meta-analysis

Figure 2 shows the individual and summary OR of the nine controlled studies, with 6/9 studies showing an OR > 1. Total OR was 2.69 (95% CI 1.68-4.30) for CMV infection in DCC attendees versus controls. There was significant heterogeneity across studies ($\chi^2/df = 8$; $I^2 = 84\%$). Sources of heterogeneity included Liu et al,³² O'Brien et al,³⁴ and Blackman et al³¹ and were attributable to differences in study recruitment methodology. In Liu et al³² and O'Brien et al,³⁴ study populations were selected randomly. In the other 28 studies, randomisation was not utilised, and controls were predominantly selected from clinic or hospital populations, with the possibility of bias towards more severe disease. The effects of high heterogeneity were reduced by using the random effects model to provide more conservative effect estimates.

3.4 | Factors contributing to higher CMV excretion rates among children attending childcare

3.4.1 | Age

Twelve studies explored the age-related prevalence of CMV excretion among children attending DCCs and identified viral excretion

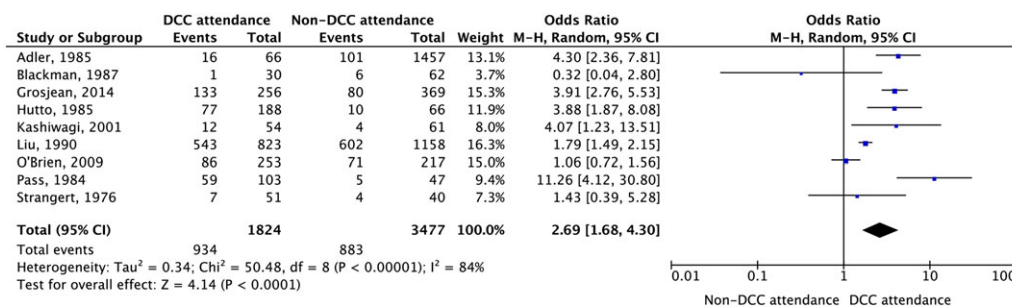


FIGURE 2 Odds ratios for CMV positivity in children attending DCCs versus controls

occurring most frequently in children under the age of 24 months. Two studies documented the viral excretion occurred most frequently in children under 12 months,^{33,51} while nine studies reported it occurred most frequently during the second year of life.^{28,37,39-42,45,48,49} Two other studies noted viral excretion occurred most frequently at age three,²⁹ and between 4 and 7 years age.³² CMV excretion rates generally declined after 2 years age.⁴⁷ Pass et al²⁸ reported that CMV excretion in a DCC increased with age, with CMV excretion found in 9% of children less than 1 year age, and 78% of children between 1 and 2 years age. The CMV excretion rate steadily declined over the following 3 years, falling to 42% by 5 years age. Hutto et al,⁴⁰ Murph et al,⁴¹ and Pass et al³⁷ similarly observed maximal viral excretion (61%-83%) occurring in children between 13 and 24 months of age, and a general decrease thereafter. The prevalence of CMV excretion by age from 0 to 49+ months is outlined in Table 3. In univariate analysis of 256 children recruited from six DCCs, every year of increase in age was associated with an 18% decrease in the risk of CMV excretion in saliva (OR 0.8, 95% CI 0.7-1.0).³⁵ Hutto et al³⁰ reported similar trends with a peak CMV excretion rate of 77% in the age group of 25 to 36 months and a third of that in those older than 48 months. Interestingly, this age-dependent variation was absent among home care children.³⁰

3.4.2 | Enrolment age

Enrolment age refers to the age of children at first DCC attendance. Adler et al⁴⁸ reported 41% (51/124) of children who first attended childcare before the age of 12 months acquired CMV, compared with 12% (10/85) of those who enrolled after the age 12 months (relative risk 3.5; 95% CI 1.9-6.5; $P < 0.001$). Similarly, Murph and Bale⁴⁴ found that CMV excretors were significantly younger at first DCC attendance than the non-excretors (5.3 ± 8.5 months versus 12.7 ± 14.8 months; $P = 0.017$). Five studies found no significant

relationship between age of first day care attendance and rate of CMV excretion.^{37,42,45,47,54}

3.4.3 | DCC size

Four studies reported an association between larger DCCs and the rate of CMV excretion in attending children,^{15,35,46,53} whereas one study revealed no association between the number of enrolled children and CMV excretion.⁴⁹

3.4.4 | Duration of DCC attendance

Two studies reported an association between duration of exposure to the DCC environment and the rate of CMV excretion.^{44,45} Volpi et al⁴⁵ found a significantly higher prevalence of CMV excretion among children who were enrolled for more than 2 years as compared with those enrolled for less than 6 months (23% versus 4%; $P = 0.028$). Children excreting CMV (excretors) were also likely to spend more time per week in the DCC (41.8 ± 9.0 hours for excretor versus 36.1 ± 10.9 hours for non-excretors; $P = 0.012$).⁴⁴ De Mello et al⁵¹ found no significant difference in prevalence of CMV excretion in relation to the time that children attended DCCs.

3.4.5 | Breastfeeding

Two studies observed that breastfeeding was related to higher CMV excretion, while one study did not find an association.^{35,45,46}

4 | DISCUSSION

This is the first systematic review of CMV infection among children in the childcare setting globally. The overall prevalence of CMV excretion was 32% (95% CI 23-41) among children attending DCCs in 28 studies with 2691 cases and 3477 controls. Comparability between studies on the Newcastle-Ottawa quality assessment scale was very good, as shown in Supporting Information Figure S2, which includes comparability of cohorts (8). In nine case-controlled studies, the rates of CMV excretion in children attending DCCs were significantly higher than those without childcare exposure, with an odds ratio of 2.69 (95% CI 1.68-4.30). Factors potentially contributing to CMV infection included younger age, with potential for increased duration of exposure, larger DCC size, close contact, duration of DCC attendance, and breastfeeding.

There are some confounders that occur with all such reviews, and some specific to CMV in DCC, none of which significantly affect our conclusions. Generally, the meta-analyses of observational studies here were affected by confounding factors inherent in the original

TABLE 3 Prevalence of CMV excretion according to age

Age (months)	1-12	13-24	25-36	37-48	49+
Pass, 1982	9%	83%	63%	23%	58%
Pass, 1984	9%	78%	-	-	42%
Hutto, 1986	28%	80%	10%	22%	-
Murph, 1986	25% ^a	61% ^b	24%	17%	0%

^a0-9 months.

^b10-24 months.

studies including (1) under-representation of developing countries; (2) some studies were performed by the same group (3/28 between 1986 and 1991^{41,44,47} and 2/28 in 1983 and 1984^{30,40}); and (3) controls were either from hospital^{29,35} or from home/out-of-home care.^{27,28,30-32} Potential confounders specific to DCCs include (1) differing size of DCC^{30,46} as crowding effects may outweigh hygiene practices as the determining factor for transmission,²⁸ and (2) differing diagnosis of CMV infection with blood PCR in 4/28 and urine or saliva excretion in 24/28 studies (Tables 1 and 2).

This review demonstrated the high prevalence of CMV infection in the DCCs and highlights the need to formulate effective prevention measures to minimise the risk of child-to-mother transmission. Measures shown and recommended to reduce CMV infection include not sharing food, utensils, cups, or washcloths with a child; regular handwashing; and not kissing a child on or near the mouth.⁵⁵⁻⁵⁹ Until an effective treatment is available, education of day-care workers and pregnant women regarding these hygiene measures remains the best preventive strategy for congenital CMV infection and subsequently congenital CMV disease. Additional measures, including further informing women, pregnant women, childcare workers, and the general population regarding the risks of congenital CMV should be considered as further ways of reducing the risk of congenital CMV.

Given the rates of CMV infection in DCCs, it is worthwhile considering highlighting prevention recommendations in DCC for using methods such as handwashing, lack of food sharing, avoiding saliva contact, and surface cleaning.⁵⁷ Many of these are usual practice, but the specific risk of CMV could be highlighted, given known lack of knowledge about this condition,¹⁸ and the primacy of this condition among serious congenital infections associated with infection of the pregnant mother.

DECLARATION OF INTEREST

None declared.

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SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section at the end of the article.

How to cite this article: Zheng QY, Huynh KT, van Zuylen WJ, Craig ME, Rawlinson WD. Cytomegalovirus infection in day care centres: A systematic review and meta-analysis of prevalence of infection in children. *Rev Med Virol*. 2018; e2011. <https://doi.org/10.1002/rmv.2011>